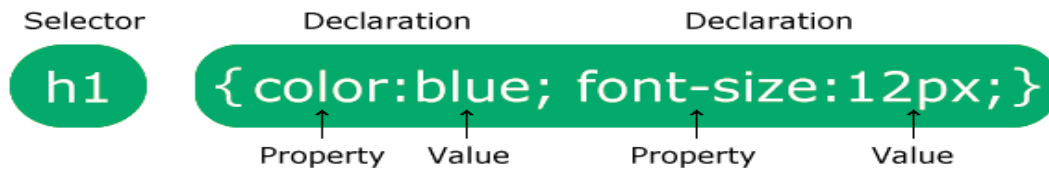


What is CSS?

- CSS stands for Cascading Style Sheets
- CSS describes how HTML elements are to be displayed on screen.
- CSS saves a lot of work. It can control the layout of multiple web pages all at once
- External stylesheets are stored in CSS files



The selector points to the HTML element you want to style.

The declaration block contains one or more declarations separated by semicolons.

Each declaration includes a CSS property name and a value, separated by a colon.

Multiple CSS declarations are separated with semicolons, and declaration blocks are surrounded by curly braces.

Types of selectors:

Element selector	P	All <p> elements
ID selector	#abc	element with id="abc"
Class selector	.abc	elements with class="abc"

```
<html>
<head>
<title>My first web</title>
<style>
  body{background-color: LightGray;}
  #title{color: Blue;}
  .hw{ color:green}
</style>
</head>
<body>
  <h1 id="title">Homework</h1>
  <em class="hw">HW0</em> is due Friday.<br/>
  <em class="hw">HW1</em> goes out Monday.<br/>
  <em>All homework due at 11:59pm.</em>
</body>
</html>
```

Homework

HW0 is due Friday.
HW1 goes out Monday.
All homework due at 11:59pm.

Three Ways to Insert CSS

There are three ways of inserting a style sheet:

- External CSS
 - Internal CSS
 - Inline CSS
-

Inline CSS

An inline style may be used to apply a unique style for a single element.

To use inline styles, add the style attribute to the relevant element. The style attribute can contain any CSS property.

```
<h1 style="color:blue;text-align:center;">This is a heading</h1>
<p style="color:red;">This is a paragraph.</p>
```

Internal CSS

An internal style sheet may be used if one single HTML page has a unique style.

The internal style is defined inside the <style> element, inside the head section.

```
<head>
<style>
body {
    background-color: maroon;
}

h1 {
    color: white;
    margin-left: 40px;
}
</style>
</head>
```

External CSS

With an external style sheet, you can change the look of an entire website by changing just one file! Each HTML page must include a reference to the external style sheet file inside the <link> element, inside the head section. The <link> element can be used in an HTML document to tell the browser where to find the CSS file used to style the page. It is an empty element (meaning it does not need a closing tag), and it

lives inside the <head> element.

```
<head>
<link rel="stylesheet" href="mystyle.css">
</head>
```

It should use three attributes:

- **href:** This specifies the path to the CSS file
- **rel:** This specifies the relationship between the HTML page and the file it is linked to. The value should be **stylesheet** when linking to a CSS file.

Note: The **link element** can be used for other purposes, for example it can be used to add an icon to your web page:

```
<link rel="icon" type="image/x-icon" href="icon.png">
```

Output:



CSS background properties

The CSS background properties are used to add background effects for elements. Such as:

- background-color
- background-image
- background-repeat
- background-attachment
- background-position

background-color:

```
<head>
<style>
body {
  background-color: rgb(200,200,200); }
h1 {
  background-color: DarkCyan;}
p {
  background-color: white;}
</style>
</head>
<body>

<h1>This is a heading</h1>
<p>This is a paragraph.</p>

</body>
```

This is a heading

This is a paragraph.

background-image

```
body {  
background-image: url("paper-texture.gif");  
}
```



CSS Borders

The border-style property specifies what kind of border to display.

```
<head>  
  <style>  
    div{border: 4px dotted black;}  
  </style>  
</head>  
<body>  
  <div>dotted</div>  
</body>
```

dotted - Defines a dotted border

dashed - Defines a dashed border

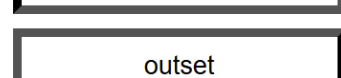
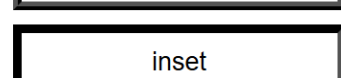
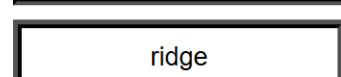
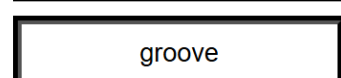
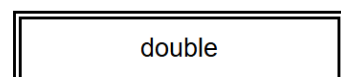
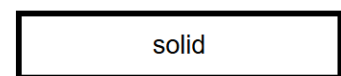
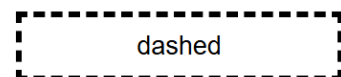
solid - Defines a solid border

double - Defines a double border

none - Defines no border

hidden - Defines a hidden border

groove, ridge, inset, outset - 3D effects, The effect depends on the border-color value



none

hidden

The border-style property can have from one to four values (for the top border, right border, bottom border, and the left border).

```

<head>
  <style>
    p{border-style: dotted dashed solid double;}
  </style>
</head>
<body>
  <p> Mixed border</p>
</body>

```

Output:

Mixed border

CSS Border Width

The border-width property specifies the width of the four borders.

The width can be set as:

- specific size (in px, pt, cm, em, etc) or by using one of the three
- pre-defined values: thin, medium, or thick

CSS Border Color

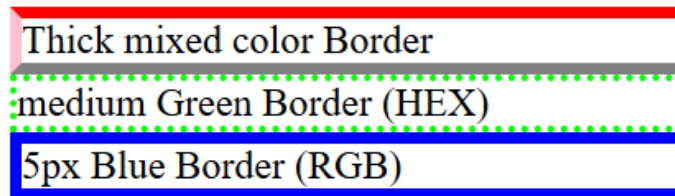
The border-color property is used to set the color of the four borders. The color can be set by:

- name - specify a color name, like "red"
- HEX - specify a HEX value, like "#ff0000"
- RGB - specify a RGB value, like "rgb(255,0,0)"

```

<head>
  <style>
    .border-mix {
      border-color: red green gray pink;
      Border-style: solid;
      border-width: thick ;
    }
    .border-hex {
      border-color: #00ff00;
      Border-style: dotted;
      border-width: medium;
    }
    .border-rgb {
      border-color: rgb(0, 0, 255);
      Border-style: solid;
      Border-width: 5px;
    }
  </style>
</head>
<body>
  <div class="border-mix">Thick mixed color Border</div>
  <div class="border-hex">medium Green Border (HEX)</div>
  <div class="border-rgb">5px Blue Border (RGB)</div>
</body>

```



Another border style property to make rounded corners is **border-radius**, for example:
border-radius :10px;

CSS Margins

The CSS margin properties are used to create space around elements, outside of any defined borders. With CSS, you have full control over the margins. There are properties for setting the margin for each side of an element (top, right, bottom, and left).

Setting Margins

You can set margins for each side of an element using:

- margin-top
- margin-right
- margin-bottom
- margin-left

```
<style>
  .margin-example {
    margin-top: 70px;
    margin-right: 150px;
    margin-bottom: 40px;
    margin-left: 50px;
    border: 2px solid black;}
</style>
</head>
<body>
  <div class="margin-example">This box has margins</div>
</body>
```

CSS Padding

The CSS padding properties are used to generate space around an element's content, inside of any defined borders. With CSS, you have full control over the padding. There are properties for setting the padding for each side of an element (top, right, bottom, and left)

Padding - Individual Sides

CSS has properties for specifying the padding for each side of an element:

- padding-top
- padding-right
- padding-bottom
- padding-left

```
<style>
.padding-example {
padding-top: 10px;
padding-right: 20px;
padding-bottom: 30px;
padding-left: 40px;
border: 2px solid black;}
</style>
</head>
<body>
  <div class="padding-example">This box has padding</div>
</body>
```



Or:

```
<head>
  <style>
    .padding-example {padding: 10px 20px 30px 40px; /* Top Right Bottom Left */}
  </style>
</head>
<body>
  <div class="padding-example">This box has padding</div>
</body>
```

Text Color

The color property is used to set the color of the text. The color is specified by:

- a color name - like "red"
- a HEX value - like "#ff0000"
- an RGB value - like "rgb(255,0,0)"

HTML Lists and CSS List Properties

The CSS list properties allow you to:

- Set different list item markers for ordered lists
- Set different list item markers for unordered lists
- Set an image as the list item marker
- Add background colors to lists and list items

```
<head>
<style>
  ul { list-style-image: url('icon.png'); }
  ol { list-style-type: upper-roman;
        background-color: lightblue;}
</style>
</head>
<body>
  <ul>
    <li>Apple</li>
    <li>Banana</li>
  </ul>
  <ol>
    <li>Step 1</li>
    <li>Step 2</li>
  </ol>
</body>
```

✓ Apple
✓ Banana

I. Step 1
II. Step 2

Block vs. Inline Elements in HTML

In HTML, elements are categorized into **block-level** and **inline** elements based on how they are displayed on the page.

1. Block Elements

A **block-level element** always **starts on a new line** and takes up the **full width** available.

- Always starts on a **new line**
- Takes up the **full width** available by default
- Can have **width, height, margin, and padding**

```
<p>This is a paragraph.</p> <!-- Block element -->
<div>This is a div.</div> <!-- Block element -->
<h1>This is a heading</h1> <!-- Block element -->
```

2. Inline Elements

An **inline element** stays **on the same line** as surrounding content and only takes up as much width as necessary.

- Does **not** start on a new line
- Only takes up as much width as necessary
- **Cannot** have width and height applied

```
<span>This is a span</span> <!-- Inline element -->
<a href="#">This is a link</a> <!-- Inline element -->
<strong>This is bold text</strong> <!-- Inline element -->
```


The **display property** in CSS determines how an HTML element behaves in the document layout. It controls how elements are displayed on a webpage.

display: block Allows an element with display: block to behave as a **block-level element**.

display: inline Allows an element with display: inline to behave as an **inline element**

inline-block is a **hybrid** between **block** and **inline** elements.

It allows an element to **stay inline**, but also **set width, height, and margins** like a block element.

CSS Layout - Align

To just center the text inside an element, use text-align: center;

CSS Layout - float and clear

The CSS float property specifies how an element should float.

The CSS clear property specifies what elements can float beside the cleared element and on which side.

CSS Tables

The border-collapse property sets whether the table borders should be collapsed into a single border:

```
<style>
table {
border-collapse: collapse;
width: 50%;
}

th, td {
border: 1px solid black;
padding: 8px;
text-align: left;
}
</style>
```

Name	Age
John	25